

Preface to the First Workshop on GenAI-XR: Generative Artificial Intelligence meets Extended Reality

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ABSTRACT

The GenAI-XR workshop aims to explore the intersection of Generative Artificial Intelligence (GenAI) and Extended Reality (XR), examining their combined potential to revolutionize various sectors including entertainment, arts, education, factory work, healthcare, architecture, and others. The workshop will provide a platform for researchers, industry professionals, and practitioners to discuss innovative methods of integrating GenAI into XR environments, enhancing immersive experiences, and personalizing interactions in real time. Through presentation and discussion sessions, participants will gain insights into the latest developments, challenges, and future directions at the intersection of GenAI and XR.

Index Terms: Generative Artificial Intelligence, Extended Reality, Artificial Intelligence and Extended Reality Integration, Personalized Interactions, Adaptive Environments, Context-Aware Systems.

1 MOTIVATION

The intersection of Generative Artificial Intelligence (GenAI) and Extended Reality (XR) represents a transformative shift in the digital landscape, with the potential to revolutionize several sectors like entertainment, education, factory work, healthcare, design, and beyond. The workshop *GenAI-XR: Generative Artificial Intelligence meets Extended Reality* (acronym: GenAI-XR) seeks to explore the innovative potential and implications of integrating these two powerful and novel technologies, focusing on both technical and practical aspects.

Recent advancements in Generative AI have demonstrated unprecedented capabilities in generating text, audio, images, 3D objects, videos, and even complete virtual environments, thus reshaping multi-modal content production by creating highly personalized, realistic, and interactive experiences.

XR represents a superset of Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) technologies that enable the blending of physical and digital worlds providing users with immersive and interactive environments with context-aware digital contents. These technologies augment human capabilities by retrieving and rendering the most relevant information in situ.

Integrating GenAI with XR not only elevates the creation of augmented environments but also introduces adaptive elements that offer new levels of personalization and user engagement. The spatial and semantic perception of XR devices (e.g., smart glasses and other wearables) can provide AI models with real-time input about

the user's current environment. Furthermore, the user's current state (mental state, heart rate, body posture, gestures, speech recognition, etc) sensed by wearable computers can also serve as valuable input to AI models.

For instance, GenAI can dynamically generate and modify 3D assets, optimize spatial configurations, and develop personalized avatars and narratives in real time, enhancing the realism and immersion of XR environments. These capabilities are particularly significant in domains such as education, where AI-generated simulations can improve learning outcomes, and healthcare, where virtual environments can aid in therapeutic interventions and surgical training. With wearable sensors, the foundational models of AI assistants can be queried with a much wider and multi-modal context rather than text prompts.

Despite the vast potential, the development of cohesive systems that merge GenAI and XR remains a complex challenge. Current research reveals a gap in modular frameworks that effectively integrate these technologies, especially in creating environments that are not only immersive but also adaptive, interactive, collaborative, and user-centric. The capacity of GenAI to autonomously produce and modify content in XR environments (and vice versa, the capacity of XR sensors to serve as input to generative models) opens new opportunities for more engaging and participative experiences. However, to fully harness these capabilities, it is crucial to design systems that enhance the adaptability and realism of XR experiences while maintaining principles that support collaborative, context-aware, and personalized interactions.

2 TOPICS OF INTEREST

The workshop focuses on the integration of GenAI technologies with XR technologies and includes but is not limited to the following topics:

- integration of GenAI models (e.g., diffusion models) with XR for dynamic contents generation (be it text, image, video, 3D models, soundscapes, etc.)
- development of techniques for enhancing user experience through AI-generated narratives, personalized avatars, virtual agents, and real-time environment adaptation
- utilizing real-time multi-modal XR (spatial computing) input to generative models for context awareness
- highlighting practical applications of GenAI-XR in healthcare, education, industrial, entertainment, arts, design, and other fields
- addressing technical challenges such as latency and computational efficiency, as well as ethical concerns related to privacy, data usage, and the authenticity of AI-generated content
- developing collaborative, multi-user XR environments powered by generative AI that support co-creation and interaction in real time.

3 ACCEPTED PAPERS AND STATISTICS

We invited novel research papers (up to 4-6 pages) and extended abstracts (up to 2 pages) about technical artifacts (demos, tools, devices, artistic installations, etc) that combine generative AI and XR.

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Submissions underwent a single-blind review process. Accepted long and short papers are archived in the adjunct conference proceedings. We also welcomed previously published works. However, previously published papers and the technical artifacts will not be archived. The submissions were handled via EasyChair.

The first edition of the workshop received a total of 11 submissions. Following the review process, the workshop organizers, taking into account the reviewers' feedback, accepted the following papers:

- Generative Artificial Intelligence Chatbot Integration for Virtual Reality Language Learning by *Long Cheng, Joy Gisler, Kenli Lao and Andreas Kunz*.
- Generative AI for Personalized Multisensory Immersive Experiences: Challenges and Opportunities for Stress Reduction by *Marilia Karla Soares Lopes and Tiago Henrique Falk*.
- Towards Supporting Literary Studies Using Virtual Reality and Generative Artificial Intelligence by *Jens Grubert, Dieter Schmalstieg and Kirsten Dickhaut*.
- RAG-VR: Leveraging Retrieval-Augmented Generation for 3D Question Answering in VR Environments by *Shiyi Ding and Ying Chen*.
- Advancing the Understanding and Evaluation of AR-Generated Scenes: When Vision-Language Models Shine and Stumble by *Lin Duan, Yanming Xiu and Maria Gorlatova*.
- From Voices to Worlds: Developing an AI-Powered Framework for 3D Object Generation in Augmented Reality by *Majid Behravan and Denis Gracanin*.

4 AGENDA AND INVITED KEYNOTE SPEAKERS

The first edition of this workshop has been scheduled for a half day on Saturday 8 March 2025 in Saint-Malo as part of VR2025, the 32nd IEEE Conference on Virtual Reality and 3D User Interfaces. The website of the workshop is available at <https://sites.google.com/view/ieeever2025-genai-xr>. The workshop features paper presentations, a panel discussion, as well as two keynote talks:

Keynote 1 Dr. Mar Gonzalez-Franco is a Computer Scientist and Neuroscientist. She is a Research Scientist Manager at Google where she leads input interactions and XRAI experience innovations for Android XR. She has envisioned Android XR multimodal and multidevice interactions to the OS and unified input vocabularies. Before working at Google, she was a Principal Researcher at Microsoft Research where she innovated for products such as Xbox, HoloLens, Soundscape and Teams, adding features used on a daily basis by millions such as Avatars for MS Teams, available to over 260 million users, that won Times Invention of the year 2022. Her technical work has produced over 40 patents (some pending), and +10 open-source projects. Apart from her technological contributions, she is a prolific scientist with over 100 publications and does regular service as program committee, chair and reviewer in top venues (ACM, IEEE, Nature Publishing, Royal Society, AAAS Science). She was awarded the IEEE VGTC VR New researcher award in 2022, and the NAE early-career engineer.

Keynote 2 Frank Steinicke is a professor of Human-Computer Interaction at the Department of Informatics at the Universität Hamburg. Before his current position, he was a professor of Computer Science in Media at the Department of Computer Science at the University of Würzburg and chair of the Immersive Media Group from 2011 to 2014. He studied Mathematics with a Minor in Computer Science at the University of Münster, from which he received his Ph.D. and Venia Legendi in Computer Science. His research interests are focused on understanding the human perceptual, cognitive, and motor abilities and limitations to improve interactions and experiences in computer-mediated realities. He received the IEEE VGTC Virtual Reality Technical Achievement Award in 2023 for his scientific contributions and was inducted into the prestigious IEEE VR Academy.

5 ORGANIZERS

The organizer team consists of experts with diverse backgrounds from augmented reality, virtual reality, AI, arts, and communications from both the industry and academia.

Pasquale Cascarano is the Primary Organizer of the workshop. He is a researcher at the University of Bologna (Italy) since July 2023, focusing on the utilization of computer science in various sectors of the creative industry (cinema, art, and fashion), as well as in the fields of medicine and biology. Research interests are particularly directed towards the study, development, and utilization of Artificial Intelligence and Extended Reality paradigms. Engaged in research projects, collaborating with various national and international public and private institutions.

Anne-Gwenn Bosser (Ecole Nationale d'Ingénieurs de Brest, Lab-STICC CNRS UMR 6285, France) is a researcher in computer science. She combines AI and HCI approaches to develop computational narratives in various media settings for sensemaking activities, narrative generation, interactive storytelling, and (embodied) conversational agents.

Jérémy Lacoche is a researcher at Orange (France) since July 2018, focusing on virtual environments and digital twins applied to the field of augmented technicians in industrial environments. His research interests concern the creation of tools for facilitating the design of collaborative immersive experiences linking the real and the virtual world with the use of various technologies such as artificial intelligence (AI) and the Internet of Things (IoT). He is engaged in standardization activities at ETSI, 3GPP and MSF and his currently chairing the Augmented Reality Framework (ARF) group at ETSI.

Shirin Hajahmadi is a Computer Science and Engineering researcher at the University of Bologna (Italy), specializing in Human-Computer Interaction and Cognitive Science. She holds degrees in computer science and cognitive psychology and completed a research internship with the HCI group at the University of Hamburg. Her work focuses on Cognitive Augmentation Systems using XR, AI, and sensor technologies to enhance cognitive functions and user experiences. Her projects include AR and VR tools for language learning, teaching complex notations, bi-manual interaction for limb differences, and real-time feedback during exercises. Shirin also explores privacy and security in XR, addressing user autonomy and ethical technology use.

Ana Stanescu is a researcher at the Graz University of Technology (Austria) since 2019, looking into AR step by step tutorials. She is interested in intelligent system assisting users learn and understand their surroundings, particularly in using machine learning in immersive environments. She is passionate about building real-time AR applications envisioning how AI can help in everyday tasks as well as industrial settings.

Gábor Sörös is a researcher in the Artificial Intelligence Research Lab of Nokia Bell Labs in Budapest (Hungary). His research interests span from wearable computer vision and spatial AI over augmented humans, augmented reality, and augmented intelligence in general, often with applications in interaction with robots and smart things. Gábor is a board member and technology lead in the Open AR Cloud EU association and is also engaged in standardization at ETSI and the Metaverse Standards Forum. Previously, he was an early contributor of 3 successful AR startups of ETH Zurich.

Besides the organizers, additional reviews were written by Valentine Bernasconi, Zsófia Gó, Csaba Józsa, Julian Moreira, Maëlic Neau, Lucchas Ribeiro Skreinig and Giacomo Vallasciani. The organisers thank all authors and reviewers for their hard work in making the workshop a success.